

DATABASE & SYSTEMS ENGINEERING

Student Attendance & Leave Approval System

Conceptual ER Design, System Scope, Requirements & Relational Schema



Problem Statement & Project Scope



The Problem

Paper registers cause data loss, attendance tampering, delayed leave approvals, and untracked class cutoffs.



In-Scope Boundaries

Session tracking, digital leave applications, multi-tier approvals, automated alerts, and makeup scheduling.



Out-of-Scope

Fee collection workflows, payroll processing, core gradebooks, and physical biometric hardware supply.

Key System Objectives

01

Automated Logging

Instant session
attendance recording.

02

Leave Automation

Paperless multi-level
application workflow.

03

Threshold Warnings

Automated alerts for low
attendance.

04

Makeup Class Sync

Seamless recovery class
scheduling.

Primary System Users & Roles



Students

View attendance logs, submit leave requests, view makeup alerts.



Faculty / Instructors

Log daily session attendance, review leaves, schedule makeup slots.



Academic Authorities (HOD / Dean)

Approve long-term leaves, monitor department compliance reports.



System Administrator

Manage users, assign section courses, configure leave policies.

Functional Requirements Matrix

 Attendance & Sessions

Session Management: Create, reschedule, or cancel class sessions.

Bulk Roll Call: Log Present, Absent, Late, or Excused status.

Threshold Calculation: Real-time percentage tracking per course section.

 Leave & Recovery

Leave Applications: Upload medical or official duty proofs.

Approval Engine: Multi-level approval and auto-status update.

Makeup Sessions: Map recovery classes to missed sessions.

Principal Data Entities (11 Core Modules)

Academic Core

- 1. Students:** Enrolled learner profiles.
- 2. Faculty:** Instructor profiles.
- 3. Courses:** Catalog subjects.
- 4. Sections:** Course offerings.

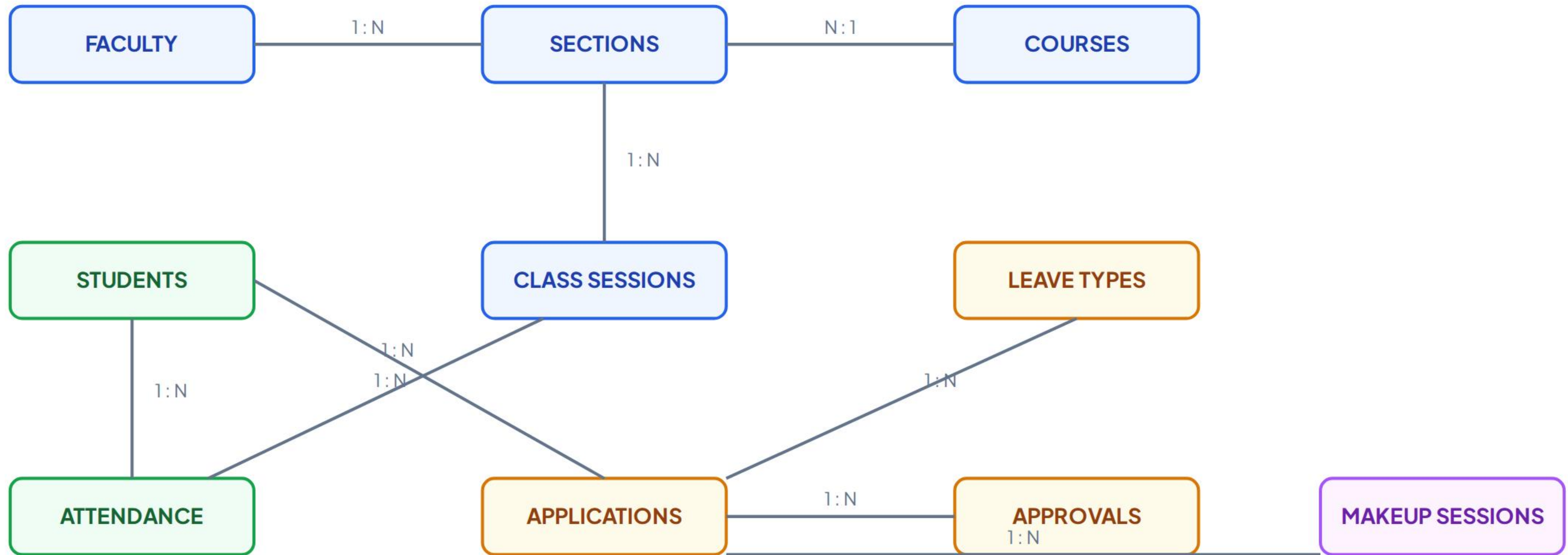
Session & Attendance

- 5. Class Sessions:** Timetabled periods.
- 6. Attendance:** Student session logs.
- 7. Makeup Sessions:** Extra recovery sessions.

Leaves & Notifications

- 8. Leave Types:** Medical, Duty, Personal.
- 9. Applications:** Leave requests.
- 10. Approvals:** Authority sign-offs.
- 11. Alerts:** System notifications.

Conceptual ER Diagram & Relationships



Relational Schema (1/2): Academic & Session Tables

STUDENTS		
PK student_id	VARCHAR(20)	Unique Roll No
first_name, last_name	VARCHAR(50)	Student Name
email, phone	VARCHAR(100)	Contact Details
FK department_id	INT	Ref DEPARTMENTS

FACULTY		
PK faculty_id	INT	Faculty Identifier
name, email	VARCHAR(100)	Instructor Info
designation	VARCHAR(50)	Professor / Asst Prof
FK department_id	INT	Ref DEPARTMENTS

COURSES		
PK course_id	VARCHAR(15)	Course Code
title, credits	VARCHAR / INT	Course Meta
min_attendance_pct	DECIMAL(5,2)	Default 75.00%

SECTIONS & CLASS_SESSIONS		
PK section_id	INT	Ref COURSES, FACULTY
PK session_id	INT	Session Primary Key
FK section_id	INT	Ref SECTIONS
session_date, room	DATE / VARCHAR	Timetable details

Relational Schema (2/2): Attendance, Leaves & Alerts

ATTENDANCE		
PK attendance_id	BIGINT	Record Primary Key
FK session_id	INT	Ref CLASS_SESSIONS
FK student_id	VARCHAR(20)	Ref STUDENTS
status	ENUM	Present, Absent, Late, Excused

APPROVALS & MAKEUP_SESSIONS		
PK approval_id	INT	Ref LEAVE_APPLICATIONS
FK approver_id	INT	Ref FACULTY
PK makeup_id	INT	Ref APPLICATION, SESSION
status / room	ENUM / VARCHAR	Approval & recovery slot

LEAVE_APPLICATIONS & TYPES		
PK application_id	INT	Application Key
FK student_id	VARCHAR(20)	Ref STUDENTS
FK leave_type_id	INT	Ref LEAVE_TYPES
start_date, end_date	DATE	Leave duration

ALERTS		
PK alert_id	BIGINT	Alert Unique ID
FK student_id	VARCHAR(20)	Ref STUDENTS
alert_type	ENUM	Low Attendance, Unexcused
triggered_at	TIMESTAMP	Automated generation timestamp



System Ready for DBMS Implementation

Normalized database structure supporting automated attendance tracking, multi-tier approvals, and instant student alerts.

3NF Normalized Schema

Automated Triggers

Role-Based Access